

Homework 1

EE-556 Mathematics of Data - Fall 2026

In this homework, we consider a multiclass classification task modeled by multinomial (softmax) logistic regression. Your goal will be to analyze the estimator and its properties (convexity, existence/uniqueness), and to derive gradients/Hessians and smoothness bounds. The first part consists of theoretical questions only.

1. Multiclass Softmax Logistic Regression - 30 Points

We now model multiclass classification with classes $c \in \{1, \dots, C\}$. For each sample $(\mathbf{a}_i, b_i)$ with $\mathbf{a}_i \in \mathbb{R}^p$ and $b_i \in \{1, \dots, C\}$, let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_C] \in \mathbb{R}^{p \times C}$ be the class weight matrix. The softmax model defines:

$$\mathbb{P}(b_i = c \mid \mathbf{a}_i) = \frac{\exp(\mathbf{a}_i^\top \mathbf{x}_c)}{\sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k)}$$

Assume i.i.d. samples $\{(\mathbf{a}_i, b_i)\}_{i=1}^n$. Our goal is to estimate $\mathbf{X}$ by maximum likelihood (and later with an ℓ_2 regularizer).

(a) (2 points)

Show that the negative log-likelihood f can be written as:

$$\begin{aligned} f(\mathbf{X}) &= -\log \mathbb{P}(b_1, \dots, b_n \mid \mathbf{a}_1, \dots, \mathbf{a}_n) \\ &= \sum_{i=1}^n \left[-\mathbf{a}_i^\top \mathbf{x}_{b_i} + \log \sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k) \right]. \end{aligned}$$

Solution: We will generally denote the column of the class weight matrix corresponding to the outcome b_i as $\mathbf{x}_{b_i}$. First, we derive the likelihood:

$$L(\mathbf{X}) = \prod_{i=1}^n \mathbb{P}(b_i \mid \mathbf{a}_i) = \prod_{i=1}^n \frac{\exp(\mathbf{a}_i^\top \mathbf{x}_{b_i})}{\sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k)}$$

Taking the log we get:

$$\log L(\mathbf{X}) = \log \left\{ \prod_{i=1}^n \frac{\exp(\mathbf{a}_i^\top \mathbf{x}_{b_i})}{\sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k)} \right\} = \sum_{i=1}^n \left[\mathbf{a}_i^\top \mathbf{x}_{b_i} - \log \sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k) \right]$$

Negating the expression gives the desired result.

(b) (3 points)

Show that $\mathbf{u} \mapsto \log\left(\sum_{k=1}^C e^{u_k}\right)$ is convex on $\mathbb{R}^C$. Then, show that $f(\mathbf{X})$ is convex. *Hint: use Jensen's inequality.*

Solution: We wish to show that $g : \mathbf{u} \mapsto \log \sum_{k=1}^C e^{u_k}$ is convex on $\mathbb{R}^C$ (where $g : \mathbb{R}^C \rightarrow \mathbb{R}$). For g to be convex on $\mathbb{R}^C$, we need:

$$g(\alpha \mathbf{u}_1 + (1 - \alpha) \mathbf{u}_2) \leq \alpha g(\mathbf{u}_1) + (1 - \alpha) g(\mathbf{u}_2)$$

for all $\mathbf{u}_1, \mathbf{u}_2 \in \mathbb{R}^C$ and for all $\alpha \in [0, 1]$.

$$\log \sum_{k=1}^C \exp(\alpha u_{1,k} + (1 - \alpha) u_{2,k}) \leq \alpha \log \sum_{k=1}^C \exp(u_{1,k}) + (1 - \alpha) \log \sum_{k=1}^C \exp(u_{2,k})$$

Exponentiating both sides:

$$\sum_{k=1}^C \exp(\alpha u_{1,k} + (1 - \alpha) u_{2,k}) \leq \exp \left\{ \log \left[\sum_{k=1}^C \exp(u_{1,k}) \right]^\alpha + \log \left[\sum_{k=1}^C \exp(u_{2,k}) \right]^{1-\alpha} \right\}$$

$$\sum_{k=1}^C \exp(\alpha u_{1,k}) \exp((1 - \alpha) u_{2,k}) \leq \left[\sum_{k=1}^C \exp(u_{1,k}) \right]^\alpha \left[\sum_{k=1}^C \exp(u_{2,k}) \right]^{1-\alpha}$$

Dividing both sides:

$$\sum_{k=1}^C \frac{\exp(\alpha u_{1,k})}{\left[\sum_{k=1}^C \exp(u_{1,k}) \right]^\alpha} \cdot \frac{\exp((1 - \alpha) u_{2,k})}{\left[\sum_{k=1}^C \exp(u_{2,k}) \right]^{1-\alpha}} \leq 1$$

Recognising the Softmax pmf, this becomes:

$$\sum_{k=1}^C \mathbb{P}(b_i = k \mid \mathbf{u}_1)^\alpha \mathbb{P}(b_i = k \mid \mathbf{u}_2)^{1-\alpha} \leq 1$$

$$\sum_{k=1}^C \mathbb{P}(b_i = k \mid \mathbf{u}_1) \frac{\mathbb{P}(b_i = k \mid \mathbf{u}_2)^{1-\alpha}}{\mathbb{P}(b_i = k \mid \mathbf{u}_1)^{1-\alpha}} \leq 1$$

Rewriting as an expectation:

$$E_{b_i \mid \mathbf{u}_1} \left\{ \left[\frac{\mathbb{P}(b_i = k \mid \mathbf{u}_2)}{\mathbb{P}(b_i = k \mid \mathbf{u}_1)} \right]^{1-\alpha} \right\} \leq 1$$

By Jensen's inequality:

$$E_{b_i \mid \mathbf{u}_1} \left\{ \left[\frac{\mathbb{P}(b_i = k \mid \mathbf{u}_2)}{\mathbb{P}(b_i = k \mid \mathbf{u}_1)} \right]^{1-\alpha} \right\} \leq \left\{ E_{b_i \mid \mathbf{u}_1} \left[\frac{\mathbb{P}(b_i = k \mid \mathbf{u}_2)}{\mathbb{P}(b_i = k \mid \mathbf{u}_1)} \right] \right\}^{1-\alpha}$$

This is because $y = x^\theta$ is concave by the definition of a convex function using the positive semi-definiteness of the Hessian (which here reduces to a scalar):

$$\frac{\partial}{\partial x} y = \theta x^{\theta-1}, \quad \frac{\partial^2}{\partial x^2} y = \theta(\theta-1)x^{\theta-2} < 0$$

for any $\theta \in [0, 1]$ (with $\theta = 1 - \alpha$). The RHS of the Jensen inequality is:

$$\begin{aligned} \left\{ E_{b_i | \mathbf{u}_1} \left[\frac{\mathbb{P}(b_i = k | \mathbf{u}_2)}{\mathbb{P}(b_i = k | \mathbf{u}_1)} \right] \right\}^{1-\alpha} &= \left(\sum_{k=1}^C \frac{\mathbb{P}(b_i = k | \mathbf{u}_2)}{\mathbb{P}(b_i = k | \mathbf{u}_1)} \mathbb{P}(b_i = k | \mathbf{u}_1) \right)^{1-\alpha} \\ &= \left(\sum_{k=1}^C \mathbb{P}(b_i = k | \mathbf{u}_2) \right)^{1-\alpha} = 1 \end{aligned}$$

since it is a pmf. This implies that:

$$E_{b_i | \mathbf{u}_1} \left\{ \left[\frac{\mathbb{P}(b_i = k | \mathbf{u}_2)}{\mathbb{P}(b_i = k | \mathbf{u}_1)} \right]^{1-\alpha} \right\} \leq 1$$

(c) (1 point)

Since the negative log-likelihood f is convex, every local minimizer is a global minimizer. The maximum likelihood estimation problem is

$$\mathbf{X}_{ML}^* = \arg \min_{\mathbf{X} \in \mathbb{R}^{p \times C}} f(\mathbf{X})$$

But does f always attain its infimum? The following three questions examine this issue.

Explain the difference between infima and minima. Give an example of a convex function on $\mathbb{R}$ that does not attain its infimum.

Let $f : \mathcal{X} \rightarrow \mathbb{R}$ be a function defined on a domain $\mathcal{X}$. The infimum of f , denoted $\inf_{x \in \mathcal{X}} f(x)$, is the greatest lower bound of its image $f(\mathcal{X})$. Formally, a real number $m = \inf_{x \in \mathcal{X}} f(x)$ if $m \leq f(x)$ for all $x \in \mathcal{X}$, and for every $\epsilon > 0$, there exists $x \in \mathcal{X}$ such that $f(x) < m + \epsilon$.

A minimum is an infimum that is explicitly attained by the function within its domain.

Specifically, the infimum is a minimum, denoted $\min_{x \in \mathcal{X}} f(x)$, if and only if $\exists x^* \in \mathcal{X}$ such that $f(x^*) = \inf_{x \in \mathcal{X}} f(x)$. Consequently, all minima are infima, but an infimum is a minimum strictly when the infimum is an element of the image set $f(\mathcal{X})$.

To illustrate a convex function on $\mathbb{R}$ that does not attain its infimum, consider $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = e^x$. This function is strictly convex and bounded below by 0. Evaluating the asymptotic behavior yields $\lim_{x \rightarrow -\infty} e^x = 0$, establishing that $\inf_{x \in \mathbb{R}} e^x = 0$. However, because $\nexists x \in \mathbb{R}$ such that $e^x = 0$, the infimum is not attained in the domain, and therefore $\min_{x \in \mathbb{R}} e^x$ is undefined.

(d) (2 points)

Assume there exists $\mathbf{X}_0 \in \mathbb{R}^{p \times C}$ such that for all i ,

$$\mathbf{a}_i^\top \mathbf{x}_{0,b_i} - \max_{k \neq b_i} \mathbf{a}_i^\top \mathbf{x}_{0,k} > 0.$$

This is called one-versus-all complete separation in multiclass settings. Give a geometric interpretation (e.g., for $p = 2$) and explain why the name is appropriate.

GEOMETRIC INTERPRETATION IS MISSING

The name "one-versus-all complete separation" is appropriate because the inequality mathematically compares the score of the correct class ("one") against the maximum score of all alternative classes ("all").

(e) (3 points)

Complete separation allows perfect classification of the training data. However, as you will show next, it prevents the existence of a finite maximum likelihood estimate.

In a one-versus-all complete separation setting (as in (d)), prove that f does not attain its minimum. Hint: consider $f(\alpha \mathbf{X}_0)$ as $\alpha \rightarrow +\infty$ and compare it to $f(\mathbf{X}_0)$.

By assumption, $\mathbf{X}_0$ achieves complete separation, that is, for all i ,

$$\mathbf{a}_i^\top \mathbf{x}_{0,b_i} - \max_{k \neq b_i} \mathbf{a}_i^\top \mathbf{x}_{0,k} > 0.$$

It is straightforward to check that if $\mathbf{X}_0$ achieves complete separation, then $\alpha \mathbf{X}_0$ achieves complete separation as well for any $\alpha > 0$:

$$\mathbf{a}_i^\top (\alpha \mathbf{x}_{0,b_i}) - \max_{k \neq b_i} \mathbf{a}_i^\top (\alpha \mathbf{x}_{0,k}) > 0 \iff \alpha \left(\mathbf{a}_i^\top \mathbf{x}_{0,b_i} - \max_{k \neq b_i} \mathbf{a}_i^\top \mathbf{x}_{0,k} \right) > 0 \iff \mathbf{a}_i^\top \mathbf{x}_{0,b_i} - \max_{k \neq b_i} \mathbf{a}_i^\top \mathbf{x}_{0,k} > 0$$

To evaluate the behavior of the negative log-likelihood under complete separation, we substitute the scaled weight matrix $\alpha \mathbf{X}_0$ into the objective function:

$$f(\alpha \mathbf{X}_0) = \sum_{i=1}^n \left[-\alpha \mathbf{a}_i^\top \mathbf{x}_{0,b_i} + \log \sum_{k=1}^C \exp(\alpha \mathbf{a}_i^\top \mathbf{x}_{0,k}) \right]$$

We can factor out the exponential term corresponding to the true class b_i from the sum inside the logarithm:

$$\begin{aligned} \sum_{k=1}^C \exp(\alpha \mathbf{a}_i^\top \mathbf{x}_{0,k}) &= \exp(\alpha \mathbf{a}_i^\top \mathbf{x}_{0,b_i}) \left(1 + \sum_{k \neq b_i} \exp(\alpha (\mathbf{a}_i^\top \mathbf{x}_{0,k} - \mathbf{a}_i^\top \mathbf{x}_{0,b_i})) \right) \\ \log \sum_{k=1}^C \exp(\alpha \mathbf{a}_i^\top \mathbf{x}_{0,k}) &= \alpha \mathbf{a}_i^\top \mathbf{x}_{0,b_i} + \log \left(1 + \sum_{k \neq b_i} \exp(\alpha (\mathbf{a}_i^\top \mathbf{x}_{0,k} - \mathbf{a}_i^\top \mathbf{x}_{0,b_i})) \right) \end{aligned}$$

Substituting this back into $f(\alpha \mathbf{X}_0)$, the $-\alpha \mathbf{a}_i^\top \mathbf{x}_{0,b_i}$ and $+\alpha \mathbf{a}_i^\top \mathbf{x}_{0,b_i}$ terms cancel out, yielding:

$$f(\alpha \mathbf{X}_0) = \sum_{i=1}^n \log \left(1 + \sum_{k \neq b_i} \exp \left(\alpha (\mathbf{a}_i^\top \mathbf{x}_{0,k} - \mathbf{a}_i^\top \mathbf{x}_{0,b_i}) \right) \right)$$

The complete separation assumption guarantees that $\alpha \mathbf{a}_i^\top \mathbf{x}_{0,b_i} > \alpha \mathbf{a}_i^\top \mathbf{x}_{0,k}$ for all $k \neq b_i$. Consequently, the exponent is strictly negative for all incorrect classes:

$$\alpha (\mathbf{a}_i^\top \mathbf{x}_{0,k} - \mathbf{a}_i^\top \mathbf{x}_{0,b_i}) < 0$$

As $\alpha \rightarrow +\infty$, the exponential of a strictly negative number approaches zero:

$$\lim_{\alpha \rightarrow +\infty} \exp \left(\alpha (\mathbf{a}_i^\top \mathbf{x}_{0,k} - \mathbf{a}_i^\top \mathbf{x}_{0,b_i}) \right) = 0$$

Evaluating the limit for the entire function gives:

$$\lim_{\alpha \rightarrow +\infty} f(\alpha \mathbf{X}_0) = \sum_{i=1}^n \log(1 + 0) = 0$$

This shows that the infimum of $f(\mathbf{X})$ is 0.

However, for any finite weight matrix $\mathbf{X} \in \mathbb{R}^{p \times C}$, the sum $\sum_{k \neq b_i} \exp(\mathbf{a}_i^\top \mathbf{x}_k - \mathbf{a}_i^\top \mathbf{x}_{b_i})$ is strictly greater than 0 because the exponential function never evaluates to zero. Therefore, $f(\mathbf{X}) > 0 \forall \mathbf{X} \in \mathbb{R}^{p \times C}$. Because the function can approach 0 asymptotically but never reach it for any finite parameters, it does not attain its minimum.

(f) (2 points)

We resolve this issue by adding a regularizer. Consider the regularized function:

$$f_\mu(\mathbf{X}) = f(\mathbf{X}) + \frac{\mu}{2} \|\mathbf{X}\|_F^2, \quad \mu > 0.$$

Show that the gradient with respect to $\mathbf{X}$ of f_μ can be expressed as:

$$\nabla_{\mathbf{X}} f_\mu(\mathbf{X}) = \sum_{i=1}^n \mathbf{a}_i (\mathbf{p}_i - \mathbf{e}_{b_i})^\top + \mu \mathbf{X},$$

where $\mathbf{e}_{b_i} \in \mathbb{R}^C$ is the one-hot vector for class b_i , $\mathbf{p}_i \in \mathbb{R}^C$ has entries $p_{i,c} = \mathbb{P}(b_i = c \mid \mathbf{a}_i)$ under the softmax model, and $\mathbf{a}_i (\mathbf{p}_i - \mathbf{e}_{b_i})^\top \in \mathbb{R}^{p \times C}$ denotes the outer product.

Solution:

By the linearity of the gradient operator, we can separate the derivative of the regularized objective function $f_\mu(\mathbf{X})$ into the gradient of the negative log-likelihood $f(\mathbf{X})$ and the gradient of the regularization term, yielding the equation:

$$\nabla_{\mathbf{X}} f_\mu(\mathbf{X}) = \nabla_{\mathbf{X}} f(\mathbf{X}) + \nabla_{\mathbf{X}} \left(\frac{\mu}{2} \|\mathbf{X}\|_F^2 \right)$$

By definition, the Frobenius norm of a matrix $\mathbf{X} \in \mathbb{R}^{p \times C}$ is given by:

$$\|\mathbf{X}\|_F = \sqrt{\sum_{i=1}^p \sum_{j=1}^C X_{ij}^2}$$

or equivalently as:

$$\|\mathbf{X}\|_F = \sqrt{\text{Tr}(\mathbf{X}^\top \mathbf{X})}$$

So:

$$\|\mathbf{X}\|_F^2 = \text{Tr}(\mathbf{X}^\top \mathbf{X})$$

Therefore, the gradient of the regularization term evaluates to:

$$\nabla_{\mathbf{X}} \left(\frac{\mu}{2} \|\mathbf{X}\|_F^2 \right) = \frac{\mu}{2} \text{Tr}(\nabla_{\mathbf{X}} \mathbf{X}^\top \mathbf{X}) = \mu \mathbf{X}$$

$\mathbf{X}$ is composed of C column vectors $\mathbf{x}_c \in \mathbb{R}^p$:

$$\mathbf{X} = [\mathbf{x}_1 \quad \mathbf{x}_2 \quad \dots \quad \mathbf{x}_C]$$

As showed in point a) the negative log-likelihood is given by:

$$f(\mathbf{X}) = \sum_{i=1}^n \left[-\mathbf{a}_i^\top \mathbf{x}_{b_i} + \log \sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k) \right]$$

And define:

$$f(\mathbf{X}) = \sum_{i=1}^n f_i(\mathbf{X})$$

where

$$f_i(\mathbf{X}) = -\mathbf{a}_i^\top \mathbf{x}_{b_i} + \log \sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k)$$

The full gradient matrix $\nabla_{\mathbf{X}} f(\mathbf{X})$ is given by:

$$\nabla_{\mathbf{X}} f(\mathbf{X}) = \begin{bmatrix} \frac{\partial f}{\partial \mathbf{x}_1} & \frac{\partial f}{\partial \mathbf{x}_2} & \dots & \frac{\partial f}{\partial \mathbf{x}_C} \end{bmatrix}$$

Where each component $\frac{\partial f}{\partial \mathbf{x}_c}$ is a p dimensional column vector containing the partial derivatives of f with respect to each scalar element of $\mathbf{x}_c$:

$$\frac{\partial f}{\partial \mathbf{x}_c} = \begin{bmatrix} \frac{\partial f}{\partial x_{1,c}} \\ \frac{\partial f}{\partial x_{2,c}} \\ \vdots \\ \frac{\partial f}{\partial x_{p,c}} \end{bmatrix}$$

The derivative of $-\mathbf{a}_i^\top \mathbf{x}_{b_i}$ with respect to $\mathbf{x}_c$ is $-\mathbf{a}_i$ if the target class b_i is equal to c , and 0

otherwise. We can write this compactly using the one-hot vector $\mathbf{e}_{b_i}$, giving:

$$\frac{\partial}{\partial \mathbf{x}_c} (-\mathbf{a}_i^\top \mathbf{x}_{b_i}) = -e_{b_i,c} \mathbf{a}_i$$

Instead using the chain rule to differentiate the summation. Because the only term in the sum that depends on $\mathbf{x}_c$ is $\exp(\mathbf{a}_i^\top \mathbf{x}_c)$, the derivative becomes:

$$\frac{\partial}{\partial \mathbf{x}_c} \log \sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k) = \frac{1}{\sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k)} \cdot \exp(\mathbf{a}_i^\top \mathbf{x}_c) \cdot \mathbf{a}_i$$

Therefore:

$$\frac{\partial}{\partial \mathbf{x}_c} \left(-\mathbf{a}_i^\top \mathbf{x}_{b_i} + \log \sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k) \right) = -e_{b_i,c} \mathbf{a}_i + \left(\frac{\exp(\mathbf{a}_i^\top \mathbf{x}_c)}{\sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k)} \right) \mathbf{a}_i$$

Note that the softmax probability for class c , denoted as $p_{i,c}$ is given by:

$$p_{i,c} = \frac{\exp(\mathbf{a}_i^\top \mathbf{x}_c)}{\sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k)}$$

Substituting $p_{i,c}$ back into the equation allows us to simplify the final gradient for sample i with respect to $\mathbf{x}_c$:

$$\frac{\partial}{\partial \mathbf{x}_c} \left(-\mathbf{a}_i^\top \mathbf{x}_{b_i} + \log \sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k) \right) = p_{i,c} \mathbf{a}_i - e_{b_i,c} \mathbf{a}_i = \mathbf{a}_i (p_{i,c} - e_{b_i,c})$$

To assemble the gradient matrix for a single sample i with respect to the entire matrix $\mathbf{X}$, we concatenate the column gradients for all classes $c \in \{1, \dots, C\}$, that is:

$$\nabla_{\mathbf{X}} f_i(\mathbf{X}) = \left[\frac{\partial}{\partial \mathbf{x}_1} \left(-\mathbf{a}_i^\top \mathbf{x}_{b_i} + \log \sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k) \right) \quad \dots \quad \frac{\partial}{\partial \mathbf{x}_C} \left(-\mathbf{a}_i^\top \mathbf{x}_{b_i} + \log \sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k) \right) \right]$$

Substituting the simplified column derivatives derived earlier, this evaluates to:

$$\begin{aligned} \nabla_{\mathbf{X}} f_i(\mathbf{X}) &= [\mathbf{a}_i(p_{i,1} - e_{b_i,1}) \quad \mathbf{a}_i(p_{i,2} - e_{b_i,2}) \quad \dots \quad \mathbf{a}_i(p_{i,C} - e_{b_i,C})] \\ &= \mathbf{a}_i [(p_{i,1} - e_{b_i,1}) \quad (p_{i,2} - e_{b_i,2}) \quad \dots \quad (p_{i,C} - e_{b_i,C})] = \mathbf{a}_i (\mathbf{p}_i - \mathbf{e}_{b_i})^\top \end{aligned}$$

By the linearity of the derivative, we sum the gradient for the single sample $\nabla_{\mathbf{X}} f_i(\mathbf{X})$ over all n independent samples to obtain the full gradient of the negative log-likelihood $f(\mathbf{X})$:

$$\nabla_{\mathbf{X}} f(\mathbf{X}) = \sum_{i=1}^n \nabla_{\mathbf{X}} f_i(\mathbf{X}) = \sum_{i=1}^n \mathbf{a}_i (\mathbf{p}_i - \mathbf{e}_{b_i})^\top$$

Next, we recall that the full regularized objective function $f_\mu(\mathbf{X})$ is the sum of this negative log-likelihood and the Frobenius norm regularization term:

$$f_\mu(\mathbf{X}) = f(\mathbf{X}) + \frac{\mu}{2} \|\mathbf{X}\|_F^2$$

Applying the gradient operator with respect to $\mathbf{X}$ to both sides gives:

$$\begin{aligned}\nabla_{\mathbf{X}} f_{\mu}(\mathbf{X}) &= \nabla_{\mathbf{X}} f(\mathbf{X}) + \nabla_{\mathbf{X}} \left(\frac{\mu}{2} \|\mathbf{X}\|_F^2 \right) \\ \nabla_{\mathbf{X}} f_{\mu}(\mathbf{X}) &= \sum_{i=1}^n \mathbf{a}_i (\mathbf{p}_i - \mathbf{e}_{b_i})^{\top} + \mu \mathbf{X}\end{aligned}$$

(g) (4 points)

Use row-wise vectorization of $\mathbf{X}$, obtained by stacking its rows one after another into a vector in $\mathbb{R}^{pC}$. Show that the Hessian of f_{μ} in these coordinates can be written as:

$$\nabla^2 f_{\mu}(\mathbf{X}) = \sum_{i=1}^n (\mathbf{a}_i \mathbf{a}_i^{\top}) \otimes \left(\text{Diag}(\mathbf{p}_i) - \mathbf{p}_i \mathbf{p}_i^{\top} \right) + \mu \mathbf{I}_{pC},$$

where $\otimes$ is the Kronecker product, and $\text{Diag}(\mathbf{p}_i) - \mathbf{p}_i \mathbf{p}_i^{\top}$ is the softmax Jacobian, which is positive semidefinite.

Solution:

Let $\mathbf{X} \in \mathbb{R}^{pC}$.

By the linearity of the Hessian operator:

$$\nabla^2 f_{\mu}(\mathbf{X}) = \nabla^2 f(\mathbf{X}) + \nabla^2 \left(\frac{\mu}{2} \|\mathbf{X}\|_F^2 \right)$$

As established in point (f), the column gradient is $\nabla_{\mathbf{X}} \left(\frac{\mu}{2} \|\mathbf{X}\|_F^2 \right) = \mu \mathbf{X}$, meaning the row gradient is $\nabla_{\mathbf{X}^{\top}} \left(\frac{\mu}{2} \|\mathbf{X}\|_F^2 \right) = \mu \mathbf{X}^{\top}$. Therefore, applying the Hessian operator yields:

$$\nabla^2 \left(\frac{\mu}{2} \|\mathbf{X}\|_F^2 \right) = \nabla_{\mathbf{X}} \nabla_{\mathbf{X}^{\top}} \left(\frac{\mu}{2} \|\mathbf{X}\|_F^2 \right) = \nabla_{\mathbf{X}} (\mu \mathbf{X}^{\top})$$

Thus, the Hessian matrix of the regularizer is simply the identity matrix scaled by μ :

$$\nabla^2 \left(\frac{\mu}{2} \|\mathbf{X}\|_F^2 \right) = \mu \mathbf{I}_{pC}$$

To derive the Hessian using only standard partial derivatives, we compute the second derivative of the negative log-likelihood for a single sample i with respect to individual scalar elements of $\mathbf{X}$.

Let X_{jc} be the scalar element in the j -th row and c -th column of $\mathbf{X}$.

From point (f):

$$f(\mathbf{X}) = \sum_{i=1}^n f_i(\mathbf{X})$$

where

$$f_i(\mathbf{X}) = -\mathbf{a}_i^\top \mathbf{x}_{b_i} + \log \sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k)$$

then the partial derivative of the negative log-likelihood f_i with respect to X_{jc} is:

$$\frac{\partial f_i}{\partial X_{jc}} = a_{i,j}(p_{i,c} - e_{b_i,c})$$

where $a_{i,j}$ is the j -th element of the feature vector $\mathbf{a}_i$.

To find the elements of the Hessian matrix, we take the second partial derivative with respect to another arbitrary element X_{kd} :

$$\frac{\partial^2 f_i}{\partial X_{jc} \partial X_{kd}} = \frac{\partial}{\partial X_{kd}} [a_{i,j}(p_{i,c} - e_{b_i,c})] = a_{i,j} \frac{\partial p_{i,c}}{\partial X_{kd}}$$

By the chain rule, we can differentiate $p_{i,c}$ with respect to X_{kd} :

$$\frac{\partial p_{i,c}}{\partial X_{kd}} = \frac{\partial}{\partial X_{kd}} \frac{\exp(\mathbf{a}_i^\top \mathbf{x}_c)}{\sum_{k=1}^C \exp(\mathbf{a}_i^\top \mathbf{x}_k)} = \sum_{m=1}^C \frac{\partial p_{i,c}}{\partial z_{i,m}} \frac{\partial z_{i,m}}{\partial X_{kd}}$$

where $z_{i,m} = \mathbf{a}_i^\top \mathbf{x}_m = \sum_{l=1}^p a_{i,l} X_{l,m}$.

Notice that $z_{i,m}$ only depends on the elements in the m -th **column** of the matrix $\mathbf{X}$. But we are differentiating with respect to X_{kd} , which is an element in the d -th **column** of $\mathbf{X}$.

So:

If $m \neq d$: The formula for $z_{i,m}$ does not contain the variable X_{kd} at all. Therefore, $\frac{\partial z_{i,m}}{\partial X_{kd}} = 0$.

If $m = d$: The formula is $z_{i,d} = \sum_{l=1}^p a_{i,l} X_{l,d}$. The derivative of this with respect to X_{kd} is just the coefficient $a_{i,k}$.

Because of this:

$$\sum_{m=1}^C \frac{\partial p_{i,c}}{\partial z_{i,m}} \frac{\partial z_{i,m}}{\partial X_{kd}} = \frac{\partial p_{i,c}}{\partial z_{i,1}}(0) + \dots + \frac{\partial p_{i,c}}{\partial z_{i,d}}(a_{i,k}) + \dots + \frac{\partial p_{i,c}}{\partial z_{i,C}}(0) = \frac{\partial p_{i,c}}{\partial z_{i,d}} a_{i,k}$$

Therefore:

$$\frac{\partial p_{i,c}}{\partial X_{kd}} = \frac{\partial p_{i,c}}{\partial z_{i,d}} a_{i,k}$$

Now we need to evaluate:

$$\frac{\partial p_{i,c}}{\partial z_{i,d}}$$

Let $c = d$

Differentiating with respect to $z_{i,c}$ directly transforms the quotient rule application into the factored probability:

$$\frac{\partial p_{i,c}}{\partial z_{i,c}} = \frac{\exp(z_{i,c}) \sum_{m=1}^C \exp(z_{i,m}) - \exp(z_{i,c})^2}{\left(\sum_{m=1}^C \exp(z_{i,m})\right)^2} = p_{i,c}(1 - p_{i,c})$$

Let $c \neq d$

Differentiating with respect to $z_{i,d}$ zeroes out the numerator derivative, reducing the quotient application directly to the negative product of the probabilities:

$$\frac{\partial p_{i,c}}{\partial z_{i,d}} = \frac{0 - \exp(z_{i,c}) \exp(z_{i,d})}{\left(\sum_{m=1}^C \exp(z_{i,m})\right)^2} = -p_{i,c}p_{i,d}$$

We can combine the two cases into a single, compact expression using δ_{cd} which evaluates to 1 if

$c = d$ and 0 if $c \neq d$:

$$\frac{\partial p_{i,c}}{\partial z_{i,d}} = p_{i,c}(\delta_{cd} - p_{i,d})$$

Substituting these back into the chain rule gives:

$$\frac{\partial p_{i,c}}{\partial X_{kd}} = p_{i,c}(\delta_{cd} - p_{i,d})a_{i,k}$$

Inserting this into our second derivative equation, we obtain the scalar element of the Hessian:

$$\frac{\partial^2 f_i}{\partial X_{jc} \partial X_{kd}} = a_{i,j}a_{i,k}p_{i,c}(\delta_{cd} - p_{i,d})$$

Notice that this scalar value is exactly the product of two distinct terms:

1. $a_{i,j}a_{i,k}$: Which is the (j, k) -th element of the $p \times p$ outer product matrix $\mathbf{a}_i \mathbf{a}_i^\top$.
2. $p_{i,c}(\delta_{cd} - p_{i,d})$, which forms the (c, d) -th entry of the $C \times C$ softmax Jacobian matrix $\text{Diag}(\mathbf{p}_i) - \mathbf{p}_i \mathbf{p}_i^\top$. This structure is immediately clear by expanding the matrix expression:

$$\text{Diag}(\mathbf{p}_i) - \mathbf{p}_i \mathbf{p}_i^\top = \begin{bmatrix} p_{i,1} & 0 & \dots & 0 \\ 0 & p_{i,2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & p_{i,C} \end{bmatrix} - \begin{bmatrix} p_{i,1}^2 & p_{i,1}p_{i,2} & \dots & p_{i,1}p_{i,C} \\ p_{i,2}p_{i,1} & p_{i,2}^2 & \dots & p_{i,2}p_{i,C} \\ \vdots & \vdots & \ddots & \vdots \\ p_{i,C}p_{i,1} & p_{i,C}p_{i,2} & \dots & p_{i,C}^2 \end{bmatrix}$$

Therefore, mapping all scalar second derivatives $\frac{\partial^2 f_i}{\partial X_{jc} \partial X_{kd}}$ into a $pC \times pC$ matrix using the Kronecker product yields:

$$\nabla^2 f_i(\mathbf{X}) = (\mathbf{a}_i \mathbf{a}_i^\top) \otimes (\text{Diag}(\mathbf{p}_i) - \mathbf{p}_i \mathbf{p}_i^\top)$$

By the linearity of the Hessian operator, we sum this over all n samples to get the Hessian of the negative log-likelihood:

$$\nabla^2 f(\mathbf{X}) = \sum_{i=1}^n (\mathbf{a}_i \mathbf{a}_i^\top) \otimes (\text{Diag}(\mathbf{p}_i) - \mathbf{p}_i \mathbf{p}_i^\top)$$

Finally, adding the Hessian of the regularization term ($\mu \mathbf{I}_{pC}$) gives the complete regularized Hessian:

$$\nabla^2 f_\mu(\mathbf{X}) = \sum_{i=1}^n (\mathbf{a}_i \mathbf{a}_i^\top) \otimes (\text{Diag}(\mathbf{p}_i) - \mathbf{p}_i \mathbf{p}_i^\top) + \mu \mathbf{I}_{pC}$$

(h) (2 points)

Show that f_μ is μ -strongly convex.

Solution:

By the lemma seen in class, a twice-differentiable convex function $f : \mathcal{Q} \rightarrow \mathbb{R}$, with $\mathcal{Q} \subseteq \mathbb{R}^{pC}$ and $f \in \mathcal{C}^2(\mathcal{Q})$, is μ -strongly convex if and only if its Hessian satisfies $\nabla^2 f(\mathbf{X}) \succeq \mu \mathbf{I}$ for all $\mathbf{X} \in \mathcal{Q}$.

From part (g), the Hessian of the regularized objective function is given by:

$$\nabla^2 f_\mu(\mathbf{X}) = \sum_{i=1}^n (\mathbf{a}_i \mathbf{a}_i^\top) \otimes (\text{Diag}(\mathbf{p}_i) - \mathbf{p}_i \mathbf{p}_i^\top) + \mu \mathbf{I}_{pC}$$

The outer product matrix $\mathbf{a}_i \mathbf{a}_i^\top$ is positive semidefinite because for any arbitrary vector $\mathbf{v}$, the quadratic form yields:

$$\mathbf{v}^\top (\mathbf{a}_i \mathbf{a}_i^\top) \mathbf{v} = (\mathbf{a}_i^\top \mathbf{v})^2 \geq 0$$

$\text{Diag}(\mathbf{p}_i) - \mathbf{p}_i \mathbf{p}_i^\top$ represents the variance-covariance matrix of a multinomial distribution that is

$\Sigma = \text{Var}(\mathbf{e}_{b_i})$, and is therefore a positive semidefinite matrix, proof of this fact is attached below.

A standard property of the Kronecker product is that the product of two positive semidefinite matrices is positive semidefinite. Consequently:

$$(\mathbf{a}_i \mathbf{a}_i^\top) \otimes (\text{Diag}(\mathbf{p}_i) - \mathbf{p}_i \mathbf{p}_i^\top) \succeq 0$$

Because the sum of positive semidefinite matrices is also positive semidefinite, summing over all n samples guarantees that the entire unregularized Hessian $\nabla^2 f(\mathbf{X})$ is positive semidefinite ($\nabla^2 f(\mathbf{X}) \succeq 0$).

Substituting this established bound back into the full Hessian expression yields:

$$\nabla^2 f_\mu(\mathbf{X}) = \nabla^2 f(\mathbf{X}) + \mu \mathbf{I}_{pC} \succeq \mu \mathbf{I}_{pC}$$

Because $\nabla^2 f_\mu(\mathbf{X}) \succeq \mu \mathbf{I}_{pC}$, the function f_μ satisfies the condition for μ -strong convexity.

Proof of $\Sigma = \text{Var}(\mathbf{e}_{b_i}) = \text{Diag}(\mathbf{p}_i) - \mathbf{p}_i \mathbf{p}_i^\top$

We represent the target label as a random vector $\mathbf{e}_{b_i} \in \mathbb{R}^C$, where $\mathbf{e}_{b_i}$ is the one-hot vector for class b_i . For any given sample, exactly one element of $\mathbf{e}_{b_i}$ is 1 and the remaining $C - 1$ elements are 0. Let $e_{i,c}$ denote the c -th component of this vector, meaning $e_{i,c} \in \{0, 1\}$.

Under our softmax model, $\mathbf{e}_{b_i}$ follows a multinomial (categorical) distribution with a single trial and probability vector $\mathbf{p}_i$.

The expected value of any component $e_{i,c}$ is the probability that $b_i = c$:

$$\mathbb{E}[e_{i,c}] = 1 \cdot p_{i,c} + 0 \cdot (1 - p_{i,c}) = p_{i,c}$$

Thus, the mean vector is $\mathbb{E}[\mathbf{e}_{b_i}] = \mathbf{p}_i$.

The variance-covariance matrix $\Sigma = \text{Var}(\mathbf{e}_{b_i})$ is populated by the variances of each component on the diagonal and the covariances between components on the off-diagonals.

For the diagonal elements:

$$\text{Var}(e_{i,c}) = \mathbb{E}[e_{i,c}^2] - (\mathbb{E}[e_{i,c}])^2 = p_{i,c} - p_{i,c}^2 = p_{i,c}(1 - p_{i,c})$$

Considering the off-diagonal elements for any two distinct classes $c \neq k$, the covariance is:

$$\text{Cov}(e_{i,c}, e_{i,k}) = \mathbb{E}[e_{i,c}e_{i,k}] - \mathbb{E}[e_{i,c}]\mathbb{E}[e_{i,k}]$$

Because the label b_i can only take one class value at a time, the vector $\mathbf{e}_{b_i}$ cannot have 1s in both position c and position k simultaneously. This means their product $e_{i,c}e_{i,k}$ is strictly 0 in all cases. Therefore, $\mathbb{E}[e_{i,c}e_{i,k}] = 0$.

Substituting the expectations yields:

$$\text{Cov}(e_{i,c}, e_{i,k}) = 0 - p_{i,c}p_{i,k} = -p_{i,c}p_{i,k}$$

Therefore:

$$\Sigma = \text{Var}(\mathbf{e}_{b_i}) = \text{Diag}(\mathbf{p}_i) - \mathbf{p}_i \mathbf{p}_i^\top$$